



Judea Pearl

ACM A.M. Turing Award 2011

For fundamental contributions to artificial intelligence through the development of a calculus for probabilistic and causal reasoning.



Daniel Kahneman

Nobel Memorial Prize in Economic Sciences 2002

For having integrated insights from psychological research into economic science, especially concerning human judgment and decision-making under uncertainty

Presidential Medal of Freedom 2013



AI101

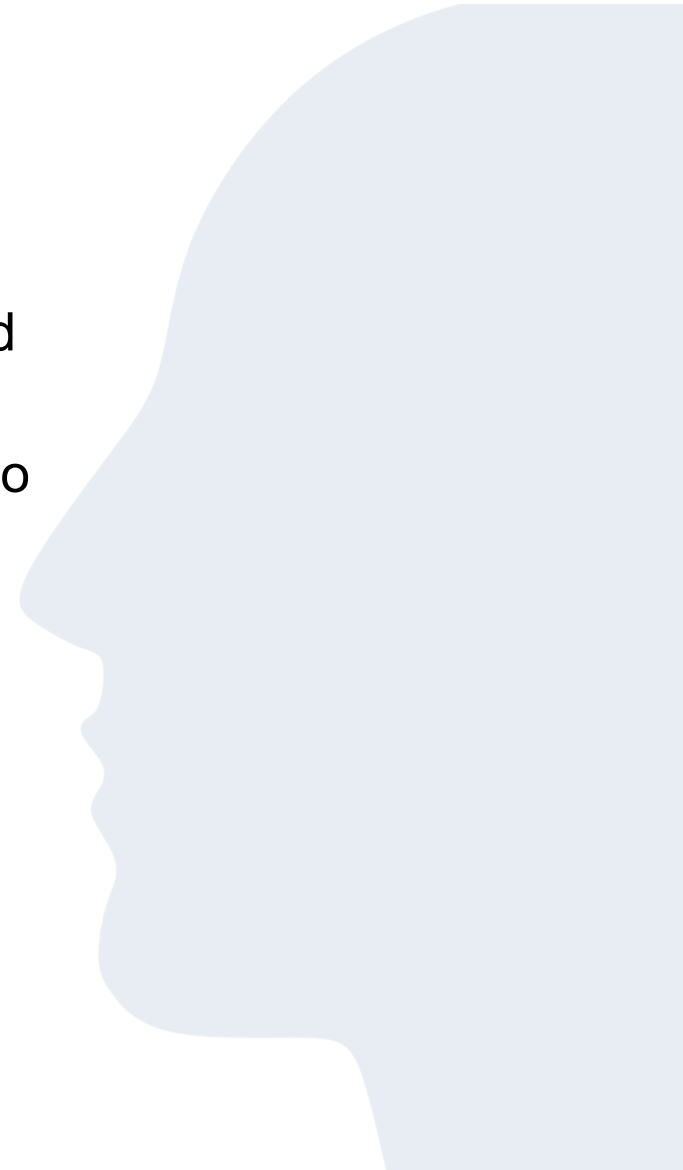
Lecture 8: Uncertainty

Recap

First Order Logic

Would like our AI to have **knowledge about the world**, and **logically draw conclusions** from it

- Notion of an object and use variables (placeholders) to specify abstract knowledge to specify true regularities
- Syntax & Semantics
- Unification
- Skolem
- Resolution
- Gödel's Incompleteness Theorem





Many situations are uncertain.
Agents have to deal with these uncertainties.



So far...

So far, agents believed that:

- (logical) statements are true or false (maybe unknown)
- actions will always do what we think they do

Unfortunately, the real world is not like that:

- Agents almost never have access to the whole truth, i.e. the complete/perfect information

Agents must deal with **uncertainty**

Uncertainty

Outline

How can agents deal with uncertainties?

Today

- Uncertainty
- Probability
- Syntax and Semantics
- Inference
- Independence and Bayes' Rule
- Bayesian Networks
- Inference in Bayesian Networks

So far...

Example: Getting to the Airport

“We want to get to the airport to take a flight. When to we leave?”

- We have different actions for getting to the airport
 - action A_t = leave for the airport t minutes before departure
- Typical problems
 - Will a given action A_t get me to the airport in time?
 - Which action is the best choice for getting me to the airport

So far...

Example: Getting to the Airport

Risks involved in the plan A_{90} will get me to the airport (leaving 90min before departure)

- partial observability (road state, other drivers' plans, etc.)
- noisy sensors (traffic reports may be wrong)
- uncertainty in action outcomes (flat tire, accident, etc.)
- immense complexity of modeling and predicting traffic

A logically correct plan: A_{90} will get me to the airport as long as my car doesn't break down, I don't run out of gas, no accident, the bridge doesn't fall down, **etc**

Unfortunately, it is impossible to model all things that can go wrong (**qualification problem**)

A more cautious plan: A_{1440} will get me to the airport

What would you do?

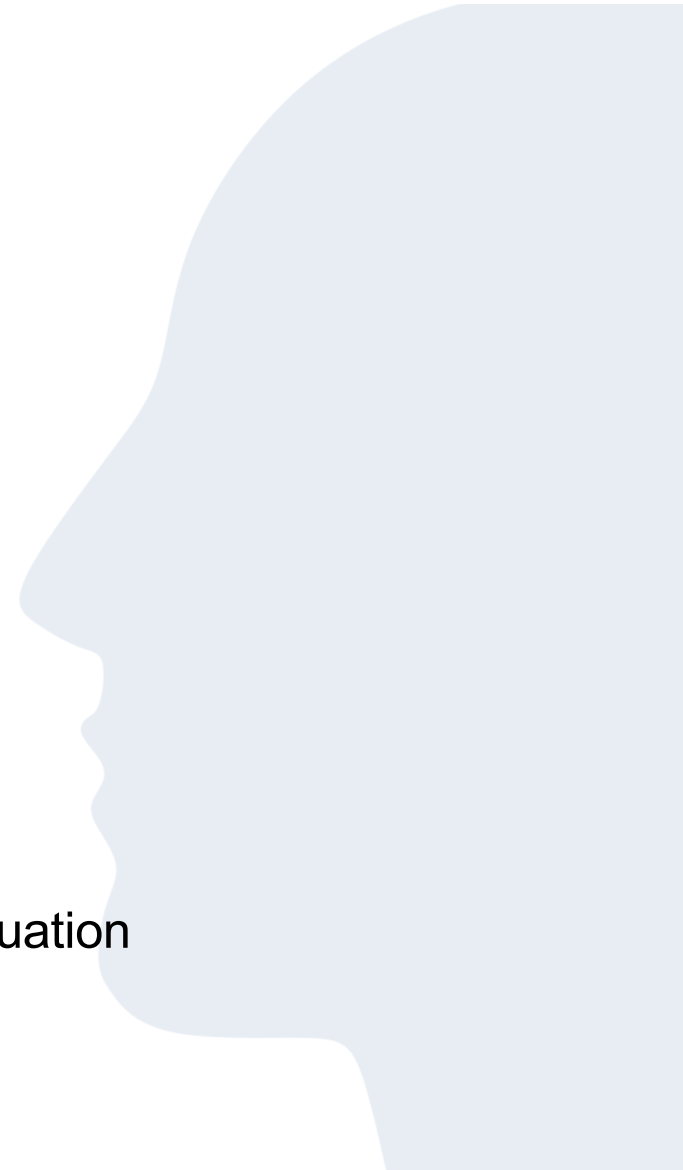
Probabilities

Probabilities are **one way** of handling uncertainty

- E.g. A_{90} will get me to the airport with probability 0.5

They **summarize the effects** that are due to

- **Laziness**
 - I don't want to list all things that must not go wrong
- **Theoretical Ignorance**
 - Some things just can't be known
 - e.g.: We cannot completely model the weather
- **Practical Ignorance**
 - Some things might not be known about the particular situation
 - e.g. Is there a traffic jam at A5?



Probabilities

How to Understand Probabilities

Probabilities are related to one's (subjective) beliefs

- A probability p means that I believe that the statement will be true in $p \cdot 100\%$ of the cases.
- E.g. There is a traffic jam in 10% of the cases
- It does not mean that the street is jammed by a degree of 10%

Probability Theory is about the **degree of belief** not the **degree of truth**

Probabilities of propositions change with new evidence:

- $P(A_{45} \text{ gets me there in time} \mid \text{no reported accidents}) = 0.06$
- $P(A_{45} \text{ gets me there in time} \mid \text{no reported accidents, 5 a.m.}) = 0.15$

Probabilities

The **degree of belief** view resolves tricky issues

Consider the probability that the sun will still exist tomorrow.

- Difficult to observe by an experiment

What is the chance that a patient has a particular disease?

- A medical doctor wants to consider other patients who are similar. But if you gather too much information to compare patients, there are no similar patients left



Probabilities

Basics

The state or **sample space** can be seen as a set of all samples:

- E.g. the sample space of a die roll is 1,2,3,4,5,6 (sides)

A **probability space** or probability model is a sample space with an assignment of probabilities per possible sample

E.g. $P(1) = P(2) = P(3) = P(4) = P(5) = P(6) = \frac{1}{6}$

An event A is any subset of the sample space:

E.g. $P(\text{roll greater 4}) = P(5) + P(6) = \frac{1}{6} + \frac{1}{6} = \frac{1}{3}$

$$\Omega$$

$$\omega \in \Omega$$

$P(\omega)$ for every $\omega \in \Omega$

$$\sum_{\omega} P(\omega) = 1$$

$$P(A) = \sum_{\omega \in A} P(\omega)$$

Probabilities

Kolmogorov's Axioms of Probability

1. All probabilities are between 0 and 1

$$0 \leq P(a) \leq 1$$

2. Necessarily true propositions have probability 1, necessarily false propositions have probability 0

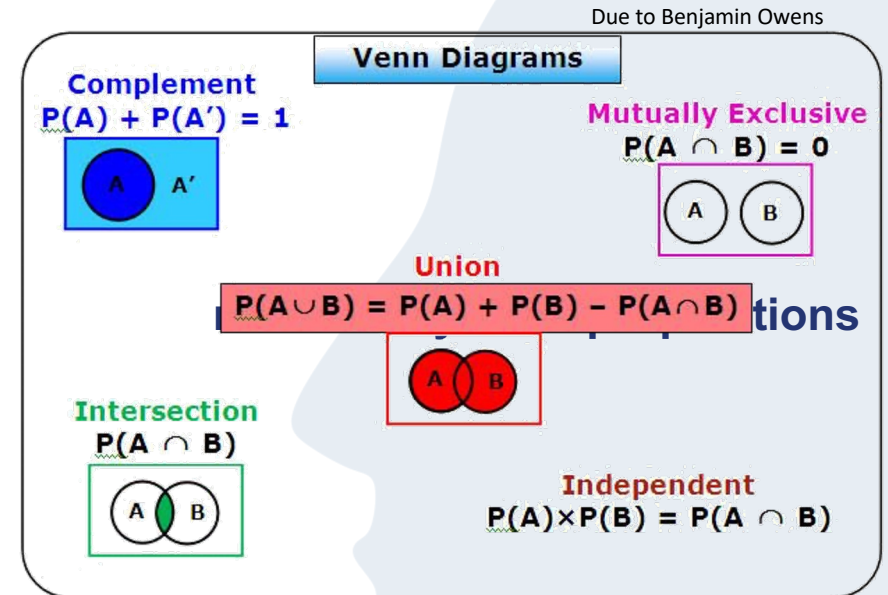
$$P(\text{false}) = 0, P(\text{true}) = 1$$

3. The probability of a disjunction is

$$P(a \vee b) = P(a) + P(b) - P(a \wedge b)$$

4. These axioms restrict the set of probabilistic beliefs that an agent can (reasonably) hold.

- similar to logical constraints like A and $\neg A$ can't both be true



Probabilities

You better do not violate the axioms of probability

Dutch Book* Theorem, Bruno de Finetti (1931)

- an agent (in the example it is Agent 1) who bets according to probabilities that violate the axioms of probability can be forced to bet so as to lose money *regardless of outcome!*

Example:

- suppose Agent 1 believes the following: $P(a) = 0.4$, $P(b) = 0.3$, $P(a \vee b) = 0.8$
- Agent 2 can now select a set of events and bet on them according to these probabilities so that she cannot loose

Agent 1		Agent 2		Outcome for Agent 1			
proposition	belief	bet	stakes	$a \wedge b$	$a \wedge \neg b$	$\neg a \wedge b$	$\neg a \wedge \neg b$
a	0.4	a	4:6	-6	-6	4	4
b	0.3	b	3:7	-7	3	-7	3
$a \vee b$	0.8	$\neg(a \vee b)$	2:8	2	2	2	-8
				-11	-1	-1	-1

*[Wikipedia] In gambling, a Dutch book is a set of odds and bets, established by the bookmaker, that ensures that the bookmaker will profit, at the expense of the gamblers, regardless of the outcome of the event (a horse race, for example) on which the gamblers bet.



Bruno de Finetti

Axioms of Probability
are violated because
 $P(a \vee b) > P(a) + P(b)$

Probabilities

Let's make our life easier: Instead of events, let us use random variables

A **random variable** is a function from atomic events to some range of values

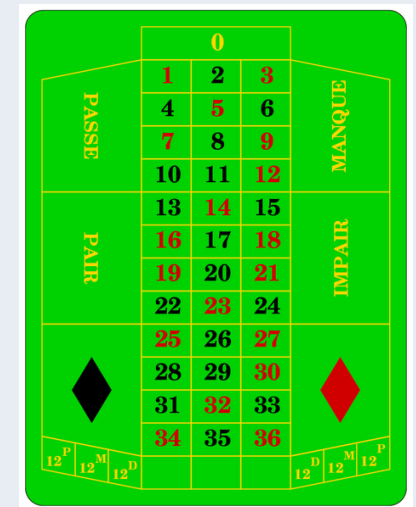
Example: Roulette

- Atomic events: numbers 0-36
- Random variables with outcomes true or false
 - Rouge | Noir, Pair | Impair, Passe | Manque
 - Transversale, Carre, Cheval
 - Douzaines premier | Milieu | Dernier
 - ...

E.g. rouge(36) = true

The probability function P over atomic events induces a **probability distribution** over all random variables X

$$P(X = x_i) = \sum_{\omega: X(\omega)=x_i} P(\omega)$$



A diagram of a roulette wheel layout on a green background. The layout is a 6x3 grid of numbers. The top row is labeled '0' in yellow. The first column is labeled 'PASSE' in yellow, and the second column is labeled 'MANQUE' in yellow. The third column is labeled 'PAIR' in yellow, and the fourth column is labeled 'IMPAIR' in yellow. The numbers are arranged as follows: Row 1: 1, 2, 3; Row 2: 4, 5, 6; Row 3: 7, 8, 9; Row 4: 10, 11, 12; Row 5: 13, 14, 15; Row 6: 16, 17, 18; Row 7: 19, 20, 21; Row 8: 22, 23, 24; Row 9: 25, 26, 27; Row 10: 28, 29, 30; Row 11: 31, 32, 33; Row 12: 34, 35, 36. The numbers 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36 are in red. The numbers 0 and 00 are in yellow. The layout is flanked by two diamond shapes, one black and one red. At the bottom, there are labels for the douzaines: '12^p', '12^M', '12^D' on the left and '12^D', '12^M', '12^p' on the right.

	0		
PASSE	1	2	3
	4	5	6
	7	8	9
	10	11	12
	13	14	15
PAIR	16	17	18
	19	20	21
	22	23	24
	25	26	27
	28	29	30
	31	32	33
	34	35	36
12 ^p 12 ^M 12 ^D			12 ^D 12 ^M 12 ^p

Probabilities

Propositions, or towards uncertain knowledge

Think of a proposition as the event where the proposition is true:

Often in AI applications, the sample points are defined by the values of a set of random variables

- i.e. the sample space is the Cartesian product of the ranges of the variables

With Boolean variables, sample points = propositional logic model

- E.g. $A=true$, $B=false$, or $a \wedge \neg b$

Proposition = disjunction of atomic events in which it is true

- E.g. $(a \vee b) \equiv (\neg a \wedge b) \vee (a \wedge \neg b) \vee (a \wedge b) \rightarrow P(a \vee b) = P(\neg a \wedge b) + P(a \wedge \neg b) + P(a \wedge b)$

Probabilities

Syntax of Propositions

Propositional or Boolean random can be true or false

- E.g. `hasUmbrella`,
- `hasUmbrella=true` is a proposition, and can be simply written as `hasUmbrella`

Discrete random variables (finite or infinite)

- E.g. `Weather` is one of $\langle \text{sunny}, \text{rain}, \text{cloudy}, \text{snow} \rangle$
- `Weather=rain` is a proposition
- Values must be exhaustive and mutually exclusive

Continuous random variables (bounded or unbounded)

- E.g. `Temp` is an unbound variable
- `Temp=25.5`, `Temp>23` are propositions

Joint Distributions

Uncertain/Quantified Truth Table

A joint distribution gives the probability of combined events

- E.g. The probability that $X=x$ and $Y=y$ is true $P(x, y) \equiv P(X = x \wedge Y = y)$

		Cancer		
		no	benigne	maligne
Smoking	no	0.768	0.024	0.008
	few	0.132	0.012	0.006
	many	0.035	0.010	0.005

The joint distribution allows us to answer any question! But how?

Marginalization (or Summing Out)

We do not want to talk always about all variable!

For any set of variables X and Y we can compute the probability

$$P(Y) = \sum_{i=1}^n P(x_i, Y)$$

The resulting distribution is called marginal distribution and its probabilities are the marginal probabilities

		Cancer		
		no	benigne	maligne
Smoking	no	0.768	0.024	0.008
	few	0.132	0.012	0.006
	many	0.035	0.010	0.005

$$P(Y = few) = P(no, few) + P(benigne, few) + P(maligne, few) = 0.15$$

Conditional Probabilities

are kind of “probability distribution for a sub-population”

A **conditional probability** can be described as the probability of $X=x$ under the assumption that $Y=y$ is true:

$$P(x|y) = \frac{P(x \wedge y)}{P(y)}$$

The **Product rule** gives us an alternative formulation

$$P(x, y) = P(x|y)P(y) = P(y|x)P(x)$$

The **Chain rule** can be derived by successive application of the Product rule:

$$\begin{aligned} P(X_1, X_2, \dots, X_n) &= P(X_1, \dots, X_{n-1})P(X_n|X_1, \dots, X_{n-1}) \\ &= P(X_1, \dots, X_{n-2})P(X_{n-1}|X_1, \dots, X_{n-2})P(X_n|X_1, \dots, X_{n-1}) \\ &= \prod_{i=1}^n P(X_i|X_1, \dots, X_{i-1}) \end{aligned}$$

Conditional Probabilities

Example

		Cancer		
		no	benigne	maligne
Smoking	no	0.768	0.024	0.008
	few	0.132	0.012	0.006
	many	0.035	0.010	0.005

Lets assume $P(Y = \text{few})$

We already know that $P(Y = \text{few}) = 0.15$

Now we want to calculate $P(X = \text{maligne} | Y = \text{few}) = \frac{P(\text{maligne}, \text{few})}{P(Y = \text{few})} = \frac{0.006}{0.15} = 0.04$

Lets assume $P(X = \text{maligne})$

We calculate $P(\text{maligne}) = P(\text{maligne}, \text{no}) + P(\text{maligne}, \text{few}) + P(\text{maligne}, \text{many}) = 0.019$

Now we want to calculate $P(Y = \text{few} | X = \text{maligne}) = \frac{P(\text{maligne}, \text{few})}{P(\text{maligne})} = \frac{0.006}{0.019} = 0.316$

Joint Distributions

Can we reduce the complexity of the joint distribution?

Yes, if we can make use of **independencies**

X and Y are independent from another if one of the following is true;

$$P(X|Y) = P(X)$$

$$P(Y|X) = P(Y)$$

$$P(X, Y) = P(X)P(Y)$$

Independent variables are not effected by the other variable

- This reduces the amount of possible values

But...

- **Absolut independent variables are rare**
- **i.e. in cancer research there are a lot of variables, none of which are independent**

Bayes Rule

In contrast to logic, no input & output (if $P > 0$)!

Product rule: $P(x, y) = P(x|y)P(y) = P(y|x)P(x)$

Bayes' rule:

Probability of the evidence,
given the belief is true.
This is called **Likelihood**

$$P(x|y) = \frac{P(y|x)P(x)}{P(y)}$$

Probability of the hypothesis
X after the evidence Y. This is
called **Posterior**

Probability of the hypothesis
before considering the evidence.
This is called **Prior**

Probability of the evidence Y
under any circumstance.
This is called **Marginalization**



Rev. Thomas Bayes (c. 1702 – 1761)
English theologian and mathematician

Bayes Rule

Example: AIDS-Test

Bayer' rule:
$$P(x|y) = \frac{P(y|x)P(x)}{P(y)}$$

Events

- Aids = a person is infected or not
- Positive = a person has a positive test result

Probabilities

$$P(\text{positive}, \text{aids}) = 0.99,$$

$$P(\text{negative}, \text{aids}) = 0.01,$$

$$P(\text{positive}, \neg \text{aids}) = 0.005,$$

$$P(\text{negative}, \neg \text{aids}) = 0.995$$

Is this test reliable?



Rev. Thomas Bayes (c. 1702 – 1761)
English theologian and mathematician

Bayes Rule

Example: AIDS-Test

Bayer' rule: $P(x|y) = \frac{P(y|x)P(x)}{P(y)}$

Probabilities

$$P(\text{positive}, \text{aids}) = 0.99,$$

$$P(\text{negative}, \text{aids}) = 0.01,$$

$$P(\text{positive}, \neg \text{aids}) = 0.005,$$

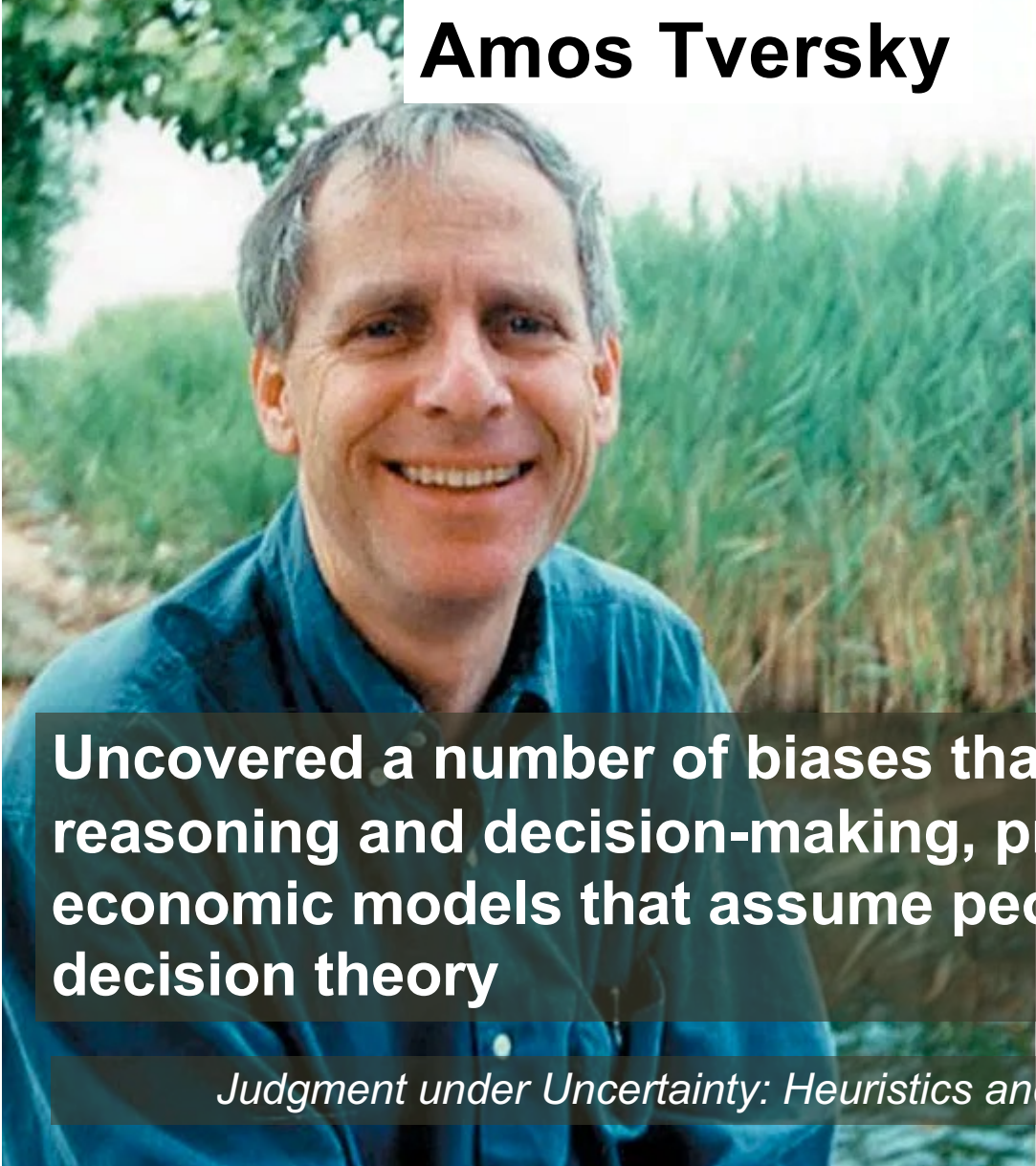
$$P(\text{negative}, \neg \text{aids}) = 0.995$$

Lets assume the risk for you to having aids is $P(\text{aids}) = 0.0001$ (Prior) and now you have a positive test result. Should you panic?

$$\begin{aligned} P(a|p) &= \frac{P(p|a)P(a)}{P(p)} = \frac{P(p|a)P(a)}{P(p|a)P(a) + P(p|\neg a)P(\neg a)} \\ &= \frac{0.99 \cdot 0.0001}{0.99 \cdot 0.0001 + 0.005 \cdot 0.9999} = \mathbf{0.0194} \end{aligned}$$



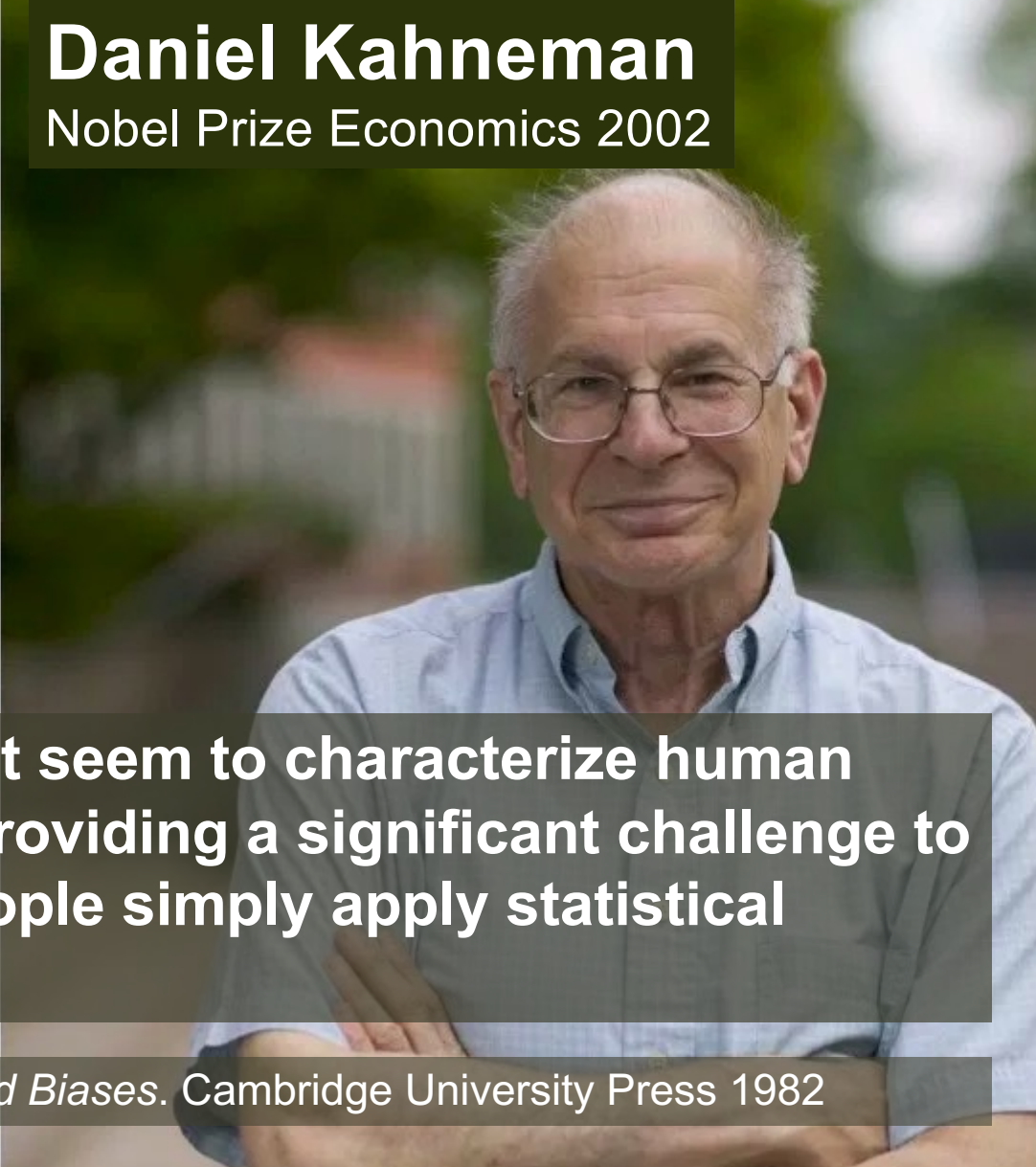
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English theologian and mathematician

A portrait of Amos Tversky, a middle-aged man with short, graying hair, smiling. He is wearing a blue button-down shirt. The background is a blurred outdoor scene with green foliage.

Amos Tversky

Uncovered a number of biases that seem to characterize human reasoning and decision-making, providing a significant challenge to economic models that assume people simply apply statistical decision theory

Judgment under Uncertainty: Heuristics and Biases. Cambridge University Press 1982

A portrait of Daniel Kahneman, an older man with glasses, wearing a light blue button-down shirt. He has a slight smile. The background is a blurred outdoor scene with green foliage.

Daniel Kahneman
Nobel Prize Economics 2002

Uncertainty in AI

How can we deal with uncertainty on a computer?

Recall Joint distribution is enumerating everything

- Worst-case run time: $O(2^n)$
 - $n = \#$ of RVs
- Space is $O(2^n)$ too
 - Size of the table of the joint distribution

Mission over? No!! Our mission has just started

Main idea: make use of independencies
to compress the representation



Summary

- Uncertainty is omnipresent
- Uncertainty can be captured using probability distributions
- Not following the axioms of probability theory makes you lose
- Marginalization, Bayes rule, chain rule, ...

You should be able to:

- Argue why not following the axioms of probabilities is bad
- Compute marginals from joint distributions

Next Week: special lecture together with Johannes “Juffi” Fürnkranz, JKU Linz, previously with TU Darmstadt

